

PIONEER Phase-I $R_{e/\mu}$ Measurement Goal

e/μ Branching Ratio

$$R_{e/\mu} = \frac{\Gamma(\pi^+ \rightarrow e^+ \nu(\gamma))}{\Gamma(\pi^+ \rightarrow \mu^+ \nu(\gamma))}$$

Target relative precision $< 0.01\%$

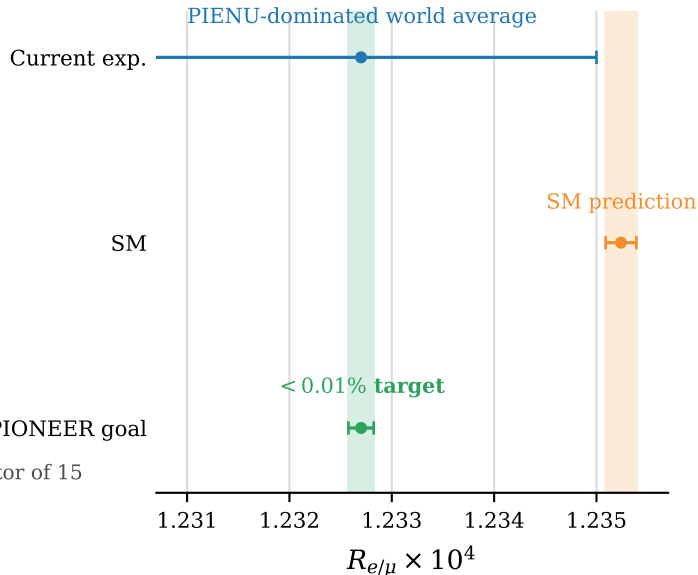
$$R_{e/\mu}^{\text{SM}} = 1.23524(15) \times 10^{-4}$$

$$R_{e/\mu}^{\text{exp}} = 1.23270(230) \times 10^{-4}$$

PIONEER goal

Theory precision exceeds current experiment by approximately a factor of 15

Factor 15 improvement



Lepton flavor universality test